

Series Practice Worksheet 4

featuring the comparison tests

For each series, circle the dominant terms in the numerator and denominator. Then determine if it converges or diverges by either using the test for divergence or comparing it with a p -series or geometric series. If you use a comparison test, tell what you're comparing to. Just get the answers here.

$$1. \sum_{n=1}^{\infty} \frac{n^2 \sqrt{n^3} + n + 1}{1 + n^2 + n^4}$$

$\sum \frac{n^{7/2}}{n^4} = \sum \frac{1}{n^{1/2}}$

$\mathcal{D}$

$$6. \sum_{n=1}^{\infty} \sqrt{\frac{n + \sin n}{n^3 + 3n - 1}}$$

$\sum \sqrt{\frac{n}{n^3}} = \sum \sqrt{\frac{1}{n^2}}$

$\mathcal{D}$

$$2. \sum_{n=1}^{\infty} \frac{2^n + 3^{n+1}}{5^{n-1} + 4^n}$$

$\sum \frac{3^{n+1}}{5^{n-1}} \leftarrow \text{geo}$

$r = \frac{3}{5}$

$\mathcal{C}$

$$7. \sum_{n=2}^{\infty} \frac{n + 4}{\ln n}$$

TFD $\lim_{n \rightarrow \infty} \frac{n+4}{\ln n} = \infty$

$\mathcal{D}$

$$3. \sum_{n=3}^{\infty} \frac{\sqrt{9n^2 + 3n - 4}}{(n-2)(n-1)n}$$

$\sum \frac{1}{n^3}$

$\sum \frac{1}{n^2}$

$\mathcal{C}$

$$8. \sum_{n=1}^{\infty} \frac{5^n + n^2}{n^3 + 2^n}$$

comp with $\sum \frac{5^n}{2^n}$

TFD $\lim_{n \rightarrow \infty} \frac{5^n + n^2}{n^3 + 2^n} = \infty$

$\mathcal{D}$

$$4. \sum_{n=1}^{\infty} \frac{1 + \frac{1}{n}}{n^2 + 2n + 1}$$

$\sum \frac{1}{n^2}$

$\mathcal{C}$

$$9. \sum_{n=1}^{\infty} \frac{7^n + 2^n}{2^{3n} + 3^n}$$

$\sum \frac{7^n}{8^n} \leftarrow \mathcal{C}$

$\uparrow$
 $(2^3)^n$
 8^n

$$5. \sum_{n=1}^{\infty} \frac{n + \arctan n}{n^2 + \arctan n}$$

$\sum \frac{n}{n^2} = \sum \frac{1}{n}$

$\mathcal{D}$

$$10. \sum_{n=1}^{\infty} \frac{(\arctan n)^n}{2^n + \sin(n)}$$

comp to $\sum \frac{(\frac{\pi}{2})^n}{2^n} \leftarrow \mathcal{C}$

1.57...

Series Practice Worksheet 5

Find the sum of the series:

1. $\sum_{n=1}^{\infty} \frac{2^n}{3^{n-2}}$

2. $\sum_{n=1}^{\infty} \frac{3^{n+2} + 4^n}{5^n}$

Sum = $\frac{a}{1-r}$

$a = \frac{3^3}{5} = \frac{27}{5}$ $r = \frac{3}{5}$

$a = \frac{4}{5}$ $r = \frac{4}{5}$

Sum = $\frac{\frac{4}{5}}{1 - \frac{4}{5}} = \frac{\frac{4}{5}}{\frac{1}{5}} = 4$

Sum = $\frac{\frac{27}{5}}{1 - \frac{3}{5}} = \frac{27}{2}$

$\frac{27}{2} + 8 = \frac{35}{2}$

Determine if each series converges or diverges. Tell which test you use. If you use a comparison test, tell what you compare it to.

3. $\sum_{n=1}^{\infty} \frac{n}{1+n^2}$

8. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+7}$

4. $\sum_{n=1}^{\infty} \frac{n+5}{1+n^3}$

9. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{1+\ln n}$ Alternating

$\lim_{n \rightarrow \infty} \frac{1}{1+\ln n} = 0$ Converge by AST

5. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt[3]{n}}$ Converge by AST

10. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2+n+1}{5-n^2}$ Diverges by oscillating

$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n}} = 0$

6. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt[3]{2}}$ Diverges by oscillating by TFD

$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{2}} = \lim_{n \rightarrow \infty} 2^{-\frac{1}{n}} = 1$

11. Use the integral test on

$\sum_{n=1}^{\infty} \frac{n}{e^{n^2}}$ $\int_1^{\infty} \frac{x}{e^{x^2}} dx = \int_1^{\infty} \frac{\frac{1}{2}}{e^u} du = \frac{1}{2} \int_1^{\infty} e^{-u} du$

$u = x^2$
 $du = 2x dx$

$\left[-\frac{1}{2} e^{-u} \right]_1^{\infty} = \left[\frac{-1}{2e^u} \right]_1^{\infty}$

7. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n+1}{10n+12}$ Alternating series

$\lim_{n \rightarrow \infty} \frac{n+1}{10n+12} = \frac{1}{10}$ D by TFD

$= \frac{-1}{2e^{\infty}} - \frac{-1}{2e^1} = \frac{1}{2e}$ Converge